

$$\begin{aligned}
\ln[*] := & f_1 := -2m(1+m)(2+s)(m+2\alpha)(1+m+2\alpha)(2m-s+2\alpha)(2+2m+s+2\alpha); \\
& f_2 := 2(2+s)(-2m+s-2\alpha)(2+2m+s+2\alpha)(12m+23m^2+10m^3-m^4+7ms+ \\
& \quad 11m^2s+4m^3s+ms^2+m^2s^2+12\alpha+46m\alpha+30m^2\alpha-4m^3\alpha+7s\alpha+22ms\alpha+12m^2s\alpha+ \\
& \quad s^2\alpha+2ms^2\alpha+30\alpha^2+32m\alpha^2-4m^2\alpha^2+16s\alpha^2+12ms\alpha^2+2s^2\alpha^2+12\alpha^3+4s\alpha^3); \\
& f_3 := -\frac{1}{2}(1+m)(1+m+2\alpha)(288m^2+1032m^3+1404m^4+840m^5+180m^6-144ms-232m^2s+ \\
& \quad 176m^3s+672m^4s+504m^5s+108m^6s-140ms^2-406m^2s^2-502m^3s^2-192m^4s^2-10ms^3- \\
& \quad 47m^2s^3-142m^3s^3-54m^4s^3+20m^5s^3+44m^2s^4+4ms^5+8m^2s^5+576m\alpha+3096m^2\alpha+5616m^3\alpha+ \\
& \quad 4200m^4\alpha+1080m^5\alpha-144s\alpha-464ms\alpha+528m^2s\alpha+2688m^3s\alpha+2520m^4s\alpha+648m^5s\alpha- \\
& \quad 140s^2\alpha-812ms^2\alpha-1506m^2s^2\alpha-768m^3s^2\alpha-10s^3\alpha-94ms^3\alpha-426m^2s^3\alpha-216m^3s^3\alpha+ \\
& \quad 20s^4\alpha+88ms^4\alpha+4s^5\alpha+16ms^5\alpha+288\alpha^2+3264m\alpha^2+8696m^2\alpha^2+8568m^3\alpha^2+2736m^4\alpha^2- \\
& \quad 320s\alpha^2+584ms\alpha^2+4188m^2s\alpha^2+5176m^3s\alpha^2+1656m^4s\alpha^2-516s^2\alpha^2-1580ms^2\alpha^2- \\
& \quad 1204m^2s^2\alpha^2-88s^3\alpha^2-452ms^3\alpha^2-344m^2s^3\alpha^2+40s^4\alpha^2+8s^5\alpha^2+1200\alpha^3+6160m\alpha^3+ \\
& \quad 8904m^2\alpha^3+3744m^3\alpha^3+232s\alpha^3+3000ms\alpha^3+5448m^2s\alpha^3+2304m^3s\alpha^3-576s^2\alpha^3- \\
& \quad 872ms^2\alpha^3-168s^3\alpha^3-256ms^3\alpha^3+1680\alpha^4+4656m\alpha^4+2896m^2\alpha^4+864s\alpha^4+2896ms\alpha^4+ \\
& \quad 1824m^2s\alpha^4-208s^2\alpha^4-64s^3\alpha^4+960\alpha^5+1184m\alpha^5+608s\alpha^5+768ms\alpha^5+192\alpha^6+128s\alpha^6); \\
& f_4 := -\frac{1}{2}(288m^2+1608m^3+3756m^4+4680m^5+3264m^6+1200m^7+180m^8-144ms+420m^2s+3476m^3s+ \\
& \quad 7288m^4s+7412m^5s+4008m^6s+1080m^7s+108m^8s-612ms^2-812m^2s^2+1774m^3s^2+ \\
& \quad 4616m^4s^2+3926m^5s^2+1500m^6s^2+216m^7s^2-932ms^3-1913m^2s^3-652m^3s^3+925m^4s^3+ \\
& \quad 758m^5s^3+162m^6s^3-689ms^4-1421m^2s^4-834m^3s^4-66m^4s^4+36m^5s^4-267ms^5- \\
& \quad 487m^2s^5-248m^3s^5-28m^4s^5-52ms^6-76m^2s^6-24m^3s^6-4ms^7-4m^2s^7+576m\alpha+ \\
& \quad 4824m^2\alpha+15024m^3\alpha+23400m^4\alpha+19584m^5\alpha+8400m^6\alpha+1440m^7\alpha-144s\alpha+840ms\alpha+ \\
& \quad 10428m^2s\alpha+29152m^3s\alpha+37060m^4s\alpha+24048m^5s\alpha+7560m^6s\alpha+864m^7s\alpha-612s^2\alpha- \\
& \quad 1624ms^2\alpha+5322m^2s^2\alpha+18464m^3s^2\alpha+19630m^4s^2\alpha+9000m^5s^2\alpha+1512m^6s^2\alpha-932s^3\alpha- \\
& \quad 3826ms^3\alpha-1956m^2s^3\alpha+3700m^3s^3\alpha+3790m^4s^3\alpha+972m^5s^3\alpha-689s^4\alpha-2842ms^4\alpha- \\
& \quad 2502m^2s^4\alpha-264m^3s^4\alpha+180m^4s^4\alpha-267s^5\alpha-974ms^5\alpha-744m^2s^5\alpha-112m^3s^5\alpha-52s^6\alpha- \\
& \quad 152ms^6\alpha-72m^2s^6\alpha-4s^7\alpha-8ms^7\alpha+288\alpha^2+4992m\alpha^2+22856m^2\alpha^2+46648m^3\alpha^2+ \\
& \quad 48200m^4\alpha^2+24600m^5\alpha^2+4896m^6\alpha^2+336s\alpha^2+11000ms\alpha^2+45052m^2s\alpha^2+74576m^3s\alpha^2+ \\
& \quad 59476m^4s\alpha^2+22216m^5s\alpha^2+2952m^6s\alpha^2-1180s^2\alpha^2+5864ms^2\alpha^2+29312m^2s^2\alpha^2+ \\
& \quad 40128m^3s^2\alpha^2+22444m^4s^2\alpha^2+4464m^5s^2\alpha^2-2502s^3\alpha^2-1864ms^3\alpha^2+6364m^2s^3\alpha^2+ \\
& \quad 8016m^3s^3\alpha^2+2464m^4s^3\alpha^2-1888s^4\alpha^2-2620ms^4\alpha^2-244m^2s^4\alpha^2+424m^3s^4\alpha^2-686s^5\alpha^2- \\
& \quad 804ms^5\alpha^2-164m^2s^5\alpha^2-120s^6\alpha^2-80ms^6\alpha^2-8s^7\alpha^2+1776\alpha^3+15664m\alpha^3+46344m^2\alpha^3+ \\
& \quad 62240m^3\alpha^3+39000m^4\alpha^3+9216m^5\alpha^3+4048s\alpha^3+31800ms\alpha^3+75488m^2s\alpha^3+77584m^3s\alpha^3+ \\
& \quad 35480m^4s\alpha^3+5616m^5s\alpha^3+2316s^2\alpha^3+21696m^2s^2\alpha^3+41864m^2s^2\alpha^3+29776m^3s^2\alpha^3+ \\
& \quad 7200m^4s^2\alpha^3-560s^3\alpha^3+5328ms^3\alpha^3+8888m^2s^3\alpha^3+3376m^3s^3\alpha^3-952s^4\alpha^3+40ms^4\alpha^3+ \\
& \quad 552m^2s^4\alpha^3-308s^5\alpha^3-104ms^5\alpha^3-32s^6\alpha^3+4080\alpha^4+22736m\alpha^4+44016m^2\alpha^4+35744m^3\alpha^4+ \\
& \quad 10384m^4\alpha^4+8728s\alpha^4+38096ms\alpha^4+55648m^2s\alpha^4+32832m^3s\alpha^4+6432m^4s\alpha^4+6512s^2\alpha^4+ \\
& \quad 22112ms^2\alpha^4+21904m^2s^2\alpha^4+6752m^3s^2\alpha^4+2000s^3\alpha^4+5120ms^3\alpha^4+2608m^2s^3\alpha^4+ \\
& \quad 192s^4\alpha^4+384ms^4\alpha^4-8s^5\alpha^4+4320\alpha^5+15776m\alpha^5+18432m^2\alpha^5+6976m^3\alpha^5+7536s\alpha^5+ \\
& \quad 20256ms\alpha^5+17056m^2s\alpha^5+4416m^3s\alpha^5+4672s^2\alpha^5+8256m^2s^2\alpha^5+3552m^2s^2\alpha^5+1216s^3\alpha^5+ \\
& \quad 1056ms^3\alpha^5+112s^4\alpha^5+2112\alpha^6+4672m\alpha^6+2560m^2\alpha^6+2720s\alpha^6+4288ms\alpha^6+1664m^2s\alpha^6+ \\
& \quad 1152s^2\alpha^6+896ms^2\alpha^6+160s^3\alpha^6+384\alpha^7+384m\alpha^7+320s\alpha^7+256ms\alpha^7+64s^2\alpha^7);
\end{aligned}$$